

11 — OSV 10 kn: the Full Five-Mode Pipeline

Proves · All five operational modes at once, and the only scenario where Level 3 contributes a visible figure.

Mechanics this scenario turns on

- Every active mode runs its **own** Level 1 — own demand, own combinations, own baseline, own optimum, own t/h. There is no single "tonnes per hour" for the vessel; the year is $\sum (\text{mode t/h} \times \text{mode hours})$.
- Level 2 and Level 3 are computed for **Transit only** (decision D4/Q5 — no workbook counterpart elsewhere). Other modes get an empty result, and the pinned-baseline radio does not reach them.
- Savings are a **difference**, so a mode whose baseline equals its optimum contributes zero — it is present on both sides and cancels, not excluded.
- `Math.Max(0, ...)` bites on short lists: with only two combinations, `2 - 3` clamps to **0** — the optimum itself — so that mode reports **zero** Level 1 savings. A battery in a mode can therefore suppress that mode's Level 1 savings; the value moves to the Battery Benefit badge instead.
- With several modes the Power Demands tables gain one row per mode, and the header is an **hours-weighted average** (total energy ÷ total hours), not a sum.
- Level 3 (DRC) damps the hotel/mission swing: `variation × 0.8`, minus whatever the battery already shaved, so the same kilowatt is never counted twice.
- When the Level 3 variation field is empty the backend looks it up from the vessel type (Bulk 250 · Container 1 500 · LNG 1 000 · otherwise the 500 default), all from `appsettings.json`.

Panels described below · Battery Contribution · Power Demands · Per-mode FOC with the actual curve · Baseline, savings, tiers · Battery Benefit

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 04.

Parametric Offshore Support vessel: curve 10 kn → **1 500 kW**, SM 15 % → effective **1 725**. Plant: ME 2×15 000 (Diesel 2-stroke Medium, id 2) · SG 2 000 · AE 4×1 000. Modes: Transit 4 000 h (hotel 220) · DP 2 360 h (thrust 3 500, hotel 300, Calm) · Port 1 200/150 · Anchor 800/200 · Maneuvering 400 (500/250). Battery 1260, Transit only. Total 8 760 h.

Battery Contribution (Transit only)

Propulsion $H = 1\,725 \times 5\% = \mathbf{86.25}$ → $J\,86.25 \times 0.35 = \mathbf{30.2}$, L **56.1** · Hotel $H = 4.4$ → $J\,0.22$, L 4.18. Tiles: **SR 60.2 / PS 30.4**. (A 1 260 kW battery against a 90.7 kW total swing — saturated 14× over.)

Power Demands (five rows)

Mode	Propulsion'	Hotel (all SG, AE=0!)	ME power	Energy kWh
Transit	1 725+56.1 = 1 781	224.2	2 005.2 (13.37 %)	8 020 970
DP	3 500	300	3 800 (25.33 %)	8 968 000
Port	0	150	150 (1 %)	180 000
Anchor	0	200	200 (1.33 %)	160 000
Maneuvering	500	250	750 (5 %)	300 000

Header: ME 17 628 970/8 760 = **2 012** · **AE = 0 for all 8 760 h** (SG-forced rule: the ME spins even in port/anchor at ~1 % load; the 4 AEs never run — observation #1 at its clearest).

Per-mode FOC with the actual curve (id 2: 5 %→175, 25 %→174.476, 30 %→172.568 ...)

Mode	SFOC(load)	FOC t/h	× hours = t
Transit	174.78 @13.37 %	0.35048	1 401.9
DP	174.35 @25.33 %	0.66252	1 563.6
Port	175.10* @1 %	0.02627	31.5
Anchor	175.10* @1.33 %	0.03502	28.0
Maneuvering	175.0 @5 %	0.13125	52.5

*extrapolated below the curve's lowest point. **IL1 total = 3 077.5 t/yr ✓**

Baseline, savings, tiers

Transit has only **2 valid combos** (SG covers the hotel → AEs idle → rejected) ⇒ the 3rd-from-worst rule clamps to index 0 = the optimal itself. Baseline **3 081.6** → savings **4.1 t/yr (0.1 %)**, negative ROI — flat SFOC + 2 near-identical rows = nothing to optimize. IL2 adds 0 (single generator carries hotel). The **Premium chip reads 36.3 t/yr** — the only tier with real value — of which the **DRC component itself is 32.2** (the other 4.1 is L1 carried over): variation 500 – 0.22 (battery) → ×0.8 → **±400**, 30 cycles/h over 8 760 h.

Battery Benefit

Covered band only 30.4 kW, Transit hours only: **21.2 t/yr = \$16 548**. A heavily oversized battery for this vessel — the model refuses to flatter it.

Takeaway: the whole pipeline end-to-end on a "boring" plant: tiny cascade, clamped baseline, zero L1/L2 value, DRC as the only earner, and the AE fleet parked by the SG-forced rule.

